

Khandaker Rafiul Islam

Full-Stack Developer · CSE Undergraduate

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SUMMARY

Computer Science undergraduate and full-stack developer who builds and ships products end to end, with a particular interest in backend systems. Most recently designed and deployed Oi Tesla Pool, a ride-pooling platform whose booking logic stays correct under concurrent requests through atomic SQL, row-level locking and a finite-state machine. Comfortable across the stack, from PostgreSQL schema design and REST APIs to responsive React interfaces, and driven by the unglamorous details that keep a system correct in production.

TECHNICAL SKILLS

Languages	Go, Python, C++, TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3
Frameworks	Node.js, Express.js, React, Next.js, React Native, Gin (Go), Tailwind CSS
Data & Backend	PostgreSQL, MySQL, MongoDB, Redis, Supabase, Firebase, RESTful APIs, JWT, Microservices
DevOps & Tools	Docker, Git, GitHub Actions, CI/CD, Linux/Bash, Postman, Vercel, Render, Jest, PyTest
Core CS	Data Structures & Algorithms, System Design, Operating Systems, Database Systems

TECHNICAL PROJECTS

Oi Tesla Pool | Direction-Aware Ride Pooling

[🔗 Live](#) | [🔗 Code](#)

Full-Stack & Backend Developer — React, TypeScript, Node.js/Express, PostgreSQL, Supabase

Sep 2026

- Built a ride-pooling platform for Dhaka's battery rickshaws: riders heading within 90° of each other share one vehicle and each pay 20% less, gated by same-pickup-zone and 3 km detour-cap rules.
- Made seat booking race-safe with a single atomic conditional UPDATE and PostgreSQL row locks (SELECT . . . FOR UPDATE), so concurrent riders can never double-book the last seat.
- Modelled every ride and pool lifecycle as a finite-state machine and enforced invariants in the database with partial unique indexes (one active ride per rider, one active pool per vehicle).
- Shipped rider and driver features: a shared pool countdown (halvable once per booking), 15-minute request expiry with bonuses, gender-only co-rider visibility for privacy, live ride tracking, ratings, and a driver console.
- Computed today and all-time driver earnings in one query (SUM . . . FILTER) backed by a partial index; stored money as integer paisa and validated every request with zod, returning DTOs only.
- Integrated Supabase Auth (email, Google and LinkedIn OAuth) with a DB trigger provisioning profiles; deployed the React/Vite frontend on Vercel, the Express API on Render and PostgreSQL on Supabase.

NestMate | Roommate Finder & Household Hub

[🔗 Live](#) | [🔗 Client](#) | [🔗 Server](#)

Full-Stack Developer — Next.js, TypeScript, Express.js, MongoDB, Firebase

May 2026

- Developed a roommate discovery platform with verified listings, favourites and direct host messaging with conversation tracking.
- Engineered a Household Hub for shared living: join-code access, member tracking and room-based organisation.
- Built an automated meal ledger and expense tracker with fair-share splitting for rent, groceries and monthly dues.
- Added real-time room chat with Firebase, and secured auth and persistence with Firebase Auth and MongoDB.

KrishiKonnnect | Agri-Tech Marketplace

[🔗 Code](#)

Mobile App Developer — React Native, Node.js, MongoDB

Oct 2023

- Engineered a cross-platform mobile app connecting farmers directly with consumers in Bangladesh.
- Developed an order management system with Node.js and MongoDB to handle concurrent transactions.
- Improved mobile performance with lazy-loaded product images and efficient state synchronisation.

EDUCATION

Jagannath University

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering · CGPA 3.25 / 4.00

2021 – Present

Relevant coursework: Algorithms, Database Systems, Software Engineering, System Design

Birshrestho Munsir Abdur Rouf Public College

Dhaka, Bangladesh

Higher Secondary Certificate (Science) · GPA 5.00 / 5.00

2019 – 2020

ACHIEVEMENTS & INTERESTS

- Competitive Programming: Active problem solver on LeetCode and Codeforces, focused on efficient algorithm design.
- Open Source: Contributed to GitHub repositories, focusing on documentation improvements and bug fixes.

- Soft Skills: Technical writing, team collaboration in Agile environments, problem solving.
- Languages: English (fluent), Bangla (native).